

LUMINOUS EFFICIENCY (lm/W)				
CANDLE 1400's	INCANDESCENT 1800's	FLUORESCENT 1920's	HID 1950's	LED 2000's
				
1	10-15	70-100	80-120	80-100

Fig. 1: Efficacy of various light sources

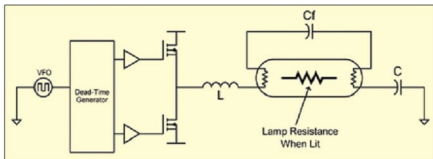


Fig. 2: Typical electronic fluorescent ballast circuit

is measured in lumens. The efficacy of several types of light sources is shown in Fig. 1.

The incandescent light bulb, hardly modified from Edison's original invention, is still widely used in residential applications. The California data showed that 59 per cent of all residential lighting energy was consumed by incandescent sources. However, the incandescent bulb offers the lowest efficacy of all sources, except for the candle.

The data drove new energy standards in the state of California, known as Title 24. The state has published compliance manuals for residential and commercial applications. To give an idea of how Title 24 is affecting the lighting industry, here is a summary of the residential requirements for new construction and remodeling:

1. High-efficacy lighting is defined as 40 lm/W for lamps less than 15W, 50 lm/W for 15W-40W lamps and 60 lm/W for lamps greater than 40W.
2. At least half of the installed wattage in kitchen must be high-efficacy. The other light sources must be

switched separately.

3. Lighting installed in garages, bathrooms, laundry rooms and utility rooms must be high-efficacy or controlled by an occupancy sensor.

4. Lighting in other rooms must be high-efficacy or controlled by an occupant sensor or dimmer.

5. Outdoor lighting must be high-efficacy or controlled by a photocell or motion sensor.

6. Ballasts for lamps over 13W should be electronic type with a switching frequency of 20 kHz or higher.

Fluorescent bulbs

Fluorescent (CFL) bulbs are highly efficient light sources, with efficacies of up to 100 lm/W. Development of the fluorescent bulb dates back to the 1800s, but Edmund Germer is given the credit for making this technology feasible in 1926. General Electric later purchased Germer's patent for fluorescent technology and brought fluorescent bulbs into widespread use by 1938.

The fluorescent bulbs work by conducting current through a tube

that is filled with mercury vapour. The vapour ionises to produce ultraviolet light. The tube has a phosphorous coating which shifts the light output to the visible spectrum.

How to drive a fluorescent bulb?

Fluorescent bulbs cannot operate directly from a power supply because of their negative impedance properties. As the current flow in the tube increases, the internal resistance decreases. This results in a run-away effect. A ballast device is required to limit the current. The ballast could be as simple as a current-limiting resistor, but it is very inefficient. Inductor ballasts have traditionally been used to control fluorescent bulb current. The inductor solution also allows an auto transformer circuit to be implemented, which steps up the line voltage to the bulb's required level.

Starting the fluorescent bulb

To start a fluorescent bulb, the input voltage must be raised to a 'strike' voltage that causes current to flow in the bulb. The strike voltage produces the initial arc in the bulb. After the strike, the resistance of the bulb rapidly decreases.

Fluorescent bulbs are available in pre-heat, rapid-start and instant-start varieties. These types of bulbs differ in the way the filaments are heated prior to striking the arc. The pre-heat type bulbs use an external starting switch to momentarily connect the filaments in series. Rapid-start bulbs use only the starting voltage to heat the filaments. Instant-start bulbs do not require filament heating and can be recognised by a single terminal at either end of the bulb. A voltage that is high enough to strike the arc without heating is applied to the bulb.

Rapid-start bulbs have the best characteristics, especially for electronic ballast applications. Though these bulbs are less efficient than other types because the filaments are continuously